



Python Function user Define function

Syntax

def \<function_name>():

```
In [ ]: print("hello")
        input()
        type()
        id()
```

hello

```
Out[ ]: 'hai'
```

```
In [16... def happy():    # func creation
        print("I\'m Happy")
        print("code reusability")
```

```
happy()  # func calling
```

I'm Happy
code reusability

```
In [3]: def happy():
        print("I\'m Happy")
        print("code reusability")
```

```
def sad():
    print("I\'m Sad")
    print("code reusability")
```

```
happy()
print()
sad()
print()
happy()
```

I'm Happy
code reusability

I'm Sad
code reusability

I'm Happy
code reusability

function with Parameters

```
In [17... def myfunc():  
    x = 100  
    y = 300  
    z = x+y  
    print("The sum of =",z)
```

```
myfunc()  
myfunc()  
myfunc()  
myfunc()
```

```
The sum of = 400  
The sum of = 400  
The sum of = 400  
The sum of = 400
```

```
In [19... def myfunc(x,y): # x,y => Parameters  
    print("X =",x)  
    print("Y =",y)  
    print("The sum of :",x+y)
```

```
myfunc(20,30) # arguments
```

```
print()  
myfunc(100,300)  
print()  
myfunc(500,0)
```

```
X = 20  
Y = 30  
The sum of : 50
```

```
X = 100  
Y = 300  
The sum of : 400
```

```
X = 500  
Y = 0  
The sum of : 500
```

function with default Parameters

```
In [32... def myfunc(x=0,y=0): # x,y => Parameters  
    print("X =",x)  
    print("Y =",y)  
    print("The sum of :",x+y)
```

```
myfunc()  
print()  
myfunc(20,10)  
print()  
myfunc(100)
```

```
print()
myfunc(1000)
```

```
X = 0
Y = 0
The sum of : 0
```

```
X = 20
Y = 10
The sum of : 30
```

```
X = 100
Y = 0
The sum of : 100
```

```
X = 1000
Y = 0
The sum of : 1000
```

return

```
In [ ]: def myfunc(x=0,y=0):
        print("X =",x)
        print("Y =",y)
        print("The sum of :",x+y)

ans = myfunc(100,200)
print(ans)
```

```
X = 100
Y = 200
The sum of : 300
None
```

```
In [40... def myfunc(x=0,y=0):
        print("X =",x)
        print("Y =",y)
        return x+y

ans = myfunc(100,200)
print("Ans =",ans)

ans2 = myfunc(40,60)

print(myfunc(400,50))

print(ans2)
```

```
X = 100
Y = 200
Ans = 300
X = 40
Y = 60
X = 400
Y = 50
450
100
```

```
In [44... def myadd(x=0,y=0):
            return x+y

            def mysub(x=0,y=0):
                return x-y

            #      80      20      = 60
            ans = myadd(50,30)-mysub(30,10)

            print("ans =",ans)

            ans = 60
```

```
In [45... def myadd(x=0,y=0):
            return x+y

            def mysub(x=0,y=0):
                return x-y

            #      70      100      30
            ans = mysub(myadd(70,30),mysub(50,20))

            print("Ans = ",ans)

            Ans = 70
```

```
In [ ]: mylist1 = [1,2,3,4,5] # 1*1=1*2=2*3=6*4=24*5=120
        mylist2 = [1,2,3,4,5,6]
        mylist3 = [1,2,3,4,5,6,7,8,9]

        720
        362880
```

```
In [ ]: mylist1 = [1,2,3,4,5]

        x = 1
        for i in mylist1:
            x = x*i
        print(x)

        120
```

```
In [58... mylist1 = [1,2,3,4,5] # 1*1=1*2=2*3=6*4=24*5=120
        mylist2 = [1,2,3,4,5,6]
```

```
mylist3 = [1,2,3,4,5,6,7,8,9]

def my_fact(mylist):
    x = 1

    for i in mylist:
        x = x*i
    return x

ans = my_fact(mylist1)
print(ans)

print(my_fact(mylist2))
print(my_fact([1,2,3,4,5,6,7,8,9]))
print(my_fact([1,2,3,4,5,6]))
print(my_fact([1,2,3,4,5,6,7,8,9.10]))
```

```
120
720
362880
720
366912.0
```

```
In [ ]: def my_fact(mylist):
        x = 1

        for i in mylist:
            x = x*i
        return x
```